Implementation Explanations

The Enrichment Analysis (Pipeline Step 10)

MH-Neuron / Modality Taxonomy project

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The enrichment analysis is the experiment that tests whether the neuron labels the pipeline produces actually mean something causally, not just statistically. Here is the whole thing in plain words, start to finish.

The starting point

By this stage of the pipeline, every neuron in the language model's feed-forward layers has already been given a label — visual, text, multimodal, or unknown — based on *when it tends to fire*: neurons that activate mostly on image-derived tokens are called visual, those that activate mostly on language tokens are called text, and so on. But firing patterns are only correlations. A neuron that fires on visual tokens might still contribute nothing to the model's visual understanding. The enrichment analysis asks the causal question: if we actually *remove* these neurons, does the model's behavior break in the way the labels predict?

The behavioral yardstick

The behavior chosen is hallucination — specifically, object hallucination measured by POPE, a benchmark of simple yes-or-no questions like "Is there a dog in this image?" asked about real photographs. A model hallucinates when it confidently says yes to an object that isn't there. The intuition is that if visual neurons genuinely carry visual evidence, then deleting them should leave the language model "blind but talkative" — it loses grounding in the image and hallucinates more. As a mirror-image control, the same experiment is run on TriviaQA, a purely textual general-knowledge quiz with no images at all. There, visual neurons should be irrelevant, and instead deleting *text* neurons should be what hurts. This two-sided design is the heart of the experiment: visual damage should implicate visual neurons, textual damage should implicate text neurons, and getting both directions right is much harder to achieve by accident than either alone.

Step one: building a trustworthy question set

Before measuring anything, the experiment cleans its measuring instrument. Models answer stochastically, so a question the model gets right one time and wrong the next would add noise to every measurement. Each candidate question is therefore asked ten times with sampling turned on, and only the questions where the model is *consistent* — always right or always wrong — are kept. This is done across all three POPE difficulty variants and capped to roughly a thousand questions, and the same filtering is applied to TriviaQA to get about a thousand clean trivia questions. The result is a stable behavioral baseline: we know exactly how the intact model performs on every retained question.

Step two: silencing neurons and measuring the damage

Now the slow part. The model's neurons are silenced in small groups — fifty at a time — by clamping their output to zero during the forward pass, while everything else in the model stays untouched. With one group silenced, the model re-answers the entire thousand-question POPE set, and its hallucination rate is compared to the intact baseline. The difference — call it delta-H — is the causal footprint of that group: a positive delta-H means those neurons were *suppressing* hallucination, so removing them made things worse. The same silenced model also re-answers the thousand TriviaQA questions, producing a second score for textual damage. Then the hooks are removed, the next fifty neurons are silenced, and the whole evaluation repeats. Since each group's score requires two thousand fresh model generations, and a single layer contains hundreds of such groups, scoring one layer takes most of a day on one GPU — which is why the cluster runs one job per layer and why this phase is where all the runs have been stuck. Each group's score is attributed equally to the fifty neurons inside it; that's an acknowledged approximation traded for tractability, since silencing neurons truly one at a time would be fifty times slower.

Step three: the enrichment test itself

Once every layer is scored, all neurons in the model are ranked by their delta-H, and the top five percent are declared the "hallucination-driving" set — the neurons whose removal most increased hallucination. Now comes the statistical question the experiment is named after: is any label category *over-represented* in that driving set relative to its share of the whole model? Suppose visual neurons make up eight percent of all neurons but eighteen percent of the driving set — that's enrichment. Formally, for each category a two-by-two table is built — driving versus not driving, in the category versus not — and Fisher's exact test gives two things: an odds ratio (how many times more likely a driving neuron is to carry that label than chance would predict, where above one means enriched and below one means depleted) and a p-value for whether the association could be a fluke. To calibrate everything, the same computation is also run a thousand times on *randomly chosen* neuron sets of the same size; these random trials should show no enrichment, providing both a sanity check and bootstrap confidence intervals. The analysis is also repeated layer by layer, producing a map of where in the network's depth the enrichment concentrates.

What success looks like

The hypothesis is a double dissociation. On the POPE side, the visual category should come out significantly enriched among hallucination-driving neurons — deleting visual neurons blinds the model. On the TriviaQA side, the text category should be the enriched one — deleting text neurons erases knowledge. Multimodal neurons may show up in both; unknown neurons and the random control should show nothing. And because the whole experiment is run twice over — once with the permutation-test labels and once with the older fixed-threshold labels — the paper can make the comparative claim that matters: the statistically principled labels predict causal function better than the heuristic ones. Every enrichment number planned for the camera-ready comes out of exactly this machinery.